

Augmented Reality Applications to Support Japanese Language Learners: A Co-Design Approach

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Abstract

In this paper, we present the results of a co-design workshop exploring the use of augmented reality (AR) to support students learning Japanese as a second language (L2). In our study, we recruited seven Japanese language instructors in Canada and explored potential applications via a co-design workshop approach. Our participants provided unique insights into the aspects of the language they most struggle to teach and collaboratively envisioned potential applications for AR to address these struggles. Based on these results, we propose four AR applications to support Japanese language learners. Implications for future work in extended reality applications in this context are also discussed.

CCS Concepts

• **Human-centered computing** → **User studies**.

Keywords

Augmented Reality, AR, Language Learning, Kanji

ACM Reference Format:

Julia M. Robinson and Victoria McArthur. 2025. Augmented Reality Applications to Support Japanese Language Learners: A Co-Design Approach. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719680>

1 Introduction

The Japanese writing system utilizes three alphabets: katakana, hiragana and kanji. Research has shown that non-native learners of the Japanese language may feel overwhelmed when first learning these alphabets, especially when learning kanji [16, 22]. In kanji, the meaning of each character in combination with other letters can drastically alter the meaning of a word depending on how specifically they are combined. Many study methods are available, including flashcards, online quizzes and writing practice sheets [33]. The ubiquity of mobile technologies has led to the development of a number of mobile games and apps designed to support kanji learning.

Many of these apps are created with augmented reality (AR), an interactive experience of a real world environment (e.g., locations, objects, people) involving the "augmentation" of these parts of the real world by computer-generated perceptual information [2]. AR differs from virtual reality or "VR" in that VR is a computer-generated experience that takes place within a simulated environment. AR experiences can be supported through a wide range of hardware, including wearable technologies, head-mounted displays (HMDs), projectors, and mobile devices. AR has been explored as a pedagogical tool in a variety of application areas, including math [7], science [32] and language learning [6]. We are interested in how AR can help students learn Japanese as a second (or subsequent) language, often denoted as L2 in the field of linguistics [22]. For over two decades, researchers have studied applications to support the learning of Japanese vocabulary (e.g., see: [35, 46]), kana (e.g., see: [35, 45, 49]), and kanji (e.g., see: [1, 15, 20, 44]). In this paper, we explore augmented reality (AR) as a tool to enhance secondary or subsequent language learning. We were further motivated to mobilize a co-design approach, working with Japanese language teachers to gain insight into the challenges of teaching the Japanese language and collaboratively envision how AR can support Japanese language learning.

2 Related Work

In this section, we present work related to Japanese language learning and an overview of past contributions related to multimedia-assisted Japanese language learning, including contributions that use AR.



Figure 1: Sample sentence in Japanese demonstrating the use of the three writing systems: katakana, hiragana, and kanji.

2.1 Japanese Language Learning

Within Japanese, katakana and hiragana are both phonetic lettering systems, writing systems where one letter corresponds to one sound. These sounds can be a vowel sound, a consonant followed

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719680>

by a vowel, or a nasal sound. Katakana is often used to write loan words (words from languages other than Japanese), for emphasis, onomatopoeia, technical and scientific terms, and foreign names. Hiragana in contrast is used for Japanese words that cannot be expressed in kanji and for grammatical inflections, such as particles. Kanji are used to write Japanese names, nouns, and stems of adjectives and verbs. Japanese particles are suffixes or short words that serve multiple grammatical purposes, including, but not limited to, denoting a subject or object, indicating possessive attributes, or connecting lists of multiple items [27]. A sample sentence demonstrating the three alphabets and particles in use is shown in Figure 1.

Gamification is quickly becoming more commonplace within language learning applications, introducing several elements to incentivize learners to interact more and learn their language of choice at a faster pace. Duolingo and similar applications use the Octalysis Framework by gamification guru Yu-Kai Chou [4]. The framework follows a set of eight "drives", to analyze human motivational forces and are utilized to build engaging and rewarding experiences for a user. The drives are as follows: Epic Meaning and Calling: The players believe that they are doing something greater than themselves. Development and Accomplishment: The product is structured for the user to feel a sense of progress towards their goals. Empowerment of Creativity and Feedback: The user can approach the product creatively and get feedback. Ownership and Possession: The user will want to manage what is owned, such as virtual goods and currencies. Social Influence and Relatedness or Envy: Relates to what Yu-Kai Chou describes as the "common and sometimes inevitable human desire to connect and compare with one another" [4], such as leaderboards. Scarcity and Impatience: Withholding something as a motivator, like expiring rewards as an engagement incentive. Unpredictability and Curiosity: The human desire to uncover what happens next, like level progression. Loss and Avoidance: The fear of losing something or having negative events transpire during the gaming process. This can relate to the loss of a daily "streak" and its associated rewards.

Each gamified application will have some level of each drive included in its design and can often work in tandem with each other to create a compelling application. Although Duolingo is credited with setting the standard for gamified language learning, it is criticized for how little it teaches Japanese kanji. Less renowned applications focus specifically on Japanese kanji, like Anki, and Benkyo by developer Romain Pellen. However, until these apps can break into mainstream, kanji learning has yet to be streamlined.

In a recent survey, Mori et al. reviewed nearly 200 publications studying language acquisition research specific to the Japanese language and found that over 80 percent of these studies are concerned with the challenges associated with learning kanji [23]. It has been suggested that students learn approximately 2000 kanji to be able to read a newspaper in Japanese [34]. Hence, non-native learners can be easily overwhelmed when first starting to learn kanji. The study of kanji involves learning each character's meaning, as well as several readings, which are classified as on yomi and kun yomi. A single kanji can have one reading or can have multiple possible readings - either on yomi or kun yomi - depending on the context in which it appears. Students may utilize any number of learning methods, including but not limited to flashcards, online quizzes and

writing practice sheets [24]. Many complicated kanji are made from a combination of common primitives to form a single character. Popular study methods encourage the use of mnemonics to help students learn common primitives and identify formed kanji, rather than individually memorizing more complex kanji (e.g., [14]). One final aspect of the Japanese language relevant to our discussion are the different politeness registers that speakers are expected to use, depending on whom they are speaking to [28]. For example, while a Japanese language teacher can address their students using informal grammar, students are expected to respond using formal grammar [19]. For situations that require more formality, the use of keigo, Japanese honorifics that are often referred to as "business Japanese", is expected [29]. In addition to register phenomena, Matsumoto and Okamoto note that the notion of a "standard" Japanese (e.g., standard dialect) is also oversimplified in Japanese language instruction [19].

2.2 Multimedia Assisted Japanese Language Learning

The ubiquity of personal computers and mobile technologies has inspired many multimedia language learning applications in the last decade. In pedagogical research, this is referred to as *Multimedia Assisted Language Learning* – a learning process which is supported by the use of various digital media such as images, sounds, video, mobile applications and serious games. Today, students learning Japanese can easily download a number of mobile apps and games (many of which are free), to support their studies. Researchers have explored hypermedia (e.g., see [25]), games (e.g., see [18, 26, 39]), interactive educational toys (e.g., see [15, 21]), computer software [36] and augmented reality [1, 12, 13, 15, 17, 20, 37, 43, 44, 48] as potential tools for motivating learners of Japanese. One of the oldest examples of multimedia assisted language learning to support Japanese involved the use of hypermedia [25]. Nakajima used Apple's Hypercard (a multimedia graphical user interface) to develop an animated tutorial program using voice digitizing to achieve an interactive, self-driven learning application [25]. Nakajima notes that a 'hypermedia approach' to kanji learning is helpful due to different information types learners must associate with a single character (e.g., stroke order, meaning, readings, etc.). Librenjak et al. [16] surveyed 32 multimedia kanji learning tools, including mobile applications (Android and iOS), interactive websites, games (e.g., *My Japanese Coach* for Nintendo DS) and websites that stream Japanese media such as dramas and anime. Some of these tools were specifically focused on kanji acquisition, while others covered various aspects of language learning (e.g., grammar, vocabulary, etc.) and enabled kanji acquisition in a more implicit manner. Of the applications designed specifically for kanji acquisition, most of these resources took the form of digital flash cards or interactive quizzes [16]. One of the more sophisticated free mobile applications currently available is *Kanji Memory Hint*, which was released for Android and iOS in 2016 by The Japan Foundation Japanese Language Institute, Kansai [11]. The application features matching games, pictures, sound, animations, kanji, stroke order and sample sentences to help students remember the kanji. A number of serious games have also been developed to support Japanese language learning [30, 39]. An early example by Stubbs explores the

design of a game-based system to support learners develop kana reading skills [39]. The game design was inspired by the Sega game *The Typing of the Dead*, a typing game presented as a first-person shooter where players must quickly and accurately type prompted phrases in order to shoot oncoming zombies. In Stubbs' version, each zombie is represented by a single kana character (hiragana or katakana) and the player must type the correct romanization in order to kill the approaching zombie. Stubbs tested the game against a traditional kana learning interface. While there were no significant differences reported between the two interfaces, Stubbs notes that participants enjoyed the gamified interface more [39]. Plecher et al. describe the design of a more complex, story-based game designed to support kanji learning [30]. In the fantasy-inspired game, dragons and humans interacted using an ancient language (kanji). Using a quest-based story mode, the player has a number of mini-games to help them practice various aspects of kanji learning, including a drawing game (practicing stroke order), a quiz game (testing knowledge of kanji meanings), a puzzle game (forming compound words) and a reading game (identifying correct readings of kanji in context). While the game was in its early stages at the time of publication, initial player feedback was positive regarding the use of mini-games to support a well-rounded approach to kanji learning [30].

2.3 Multimedia Assisted Japanese Language Learning with Augmented Reality

One of the first applications of AR for learning kanji was developed by Wagner and Barakonyi [44] – a two-player kanji matching game developed for PDAs in which opponents take turns trying to match ten kanji cards to vocabulary. Since then, more than a dozen AR applications have been developed, with most focusing on aspects of the written language, including kana (e.g., see: [35, 45, 49]) and kanji (e.g., see: [1, 12, 13, 15, 17, 20, 37, 43, 48]). We present a summary of our review of these papers in the figure below. Unsurprisingly, the majority of applications focused on various aspects of kanji, aligning with the trends noted in the literature survey on research featuring Japanese language learners [23]. Applications were commonly developed for mobile devices and featured common pedagogical objects such as flash cards or cue cards with AR (e.g., see: [1, 20, 48]). Notable exceptions included the use of 3D printed objects to explore tangible AR (T-AR) to support kanji learning through touch [12, 15]. Lin et al. [17] evaluated a collaborative tabletop AR game for two players using marker-based AR to track participants selections of kanji components or radicals. If participants placed radicals on the table that collectively make a kanji, participants were prompted to collaboratively create a short story to help them remember the kanji's meaning. The design of the application was motivated by the Heisig method described above [14]. Nearly all of the applications summarized in Figure 1 explored the augmentation of custom content (e.g., custom 3D printed objects, custom kanji flashcards, etc.) or explored the juxtaposition of AR content in physical spaces [37, 43].

However, one common gap in the previously described literature was that many of the proposed application designs were not produced under the consultation of experts in Japanese language pedagogy. With the exception of Shintaku's AR application [37],

none of the authors from our sample reported designing applications based on consultation with Japanese language teachers. This gap was the motivation for our study design: to mobilize co-design methodology in exploring potential AR applications to support Japanese language learners. We believe none of the paper authors from our sample had training in Japanese language pedagogy – the majority were from disciplines relevant to AR software development. In turn, this can limit the efficacy of those apps.

3 Methodology

We mobilized a co-design approach to see what kinds of prototypes would emerge through collaborative ideation with Japanese language instructors. Co-design is a collaborative methodology that engages stakeholders at key points in the iterative design framework [41]. While stakeholders generally refer to targeted user communities, researchers have also demonstrated the benefits of recruiting domain experts in co-design studies to bridge knowledge gaps between domain experts, designers and researchers (e.g., see: [47]). We recruited participants for our study via email and snowball sampling from Japanese language courses offered at local Ottawa Universities, the Embassy of Japan in Ottawa, and Global Affairs Canada. At the beginning of the workshop, participants filled out a short survey that included demographic questions, their experience teaching Japanese and struggles their students face while learning Japanese. Seven participants agreed to participate in our workshop. Six identified as female and one identified as male. Participants were aged 23 to 72 ($M = 51.7$, $SD = 15.7$). We structured our workshop in three phases: introduction, ideation, and discussion. In the introduction phase, we introduced AR to our participants, providing an overview of some of the different types of AR (augmented books, location-based AR, etc.), the different ways to access AR content (e.g., smartphone, tablet, HMD, etc.) and described several examples of previous AR projects in a variety of educational contexts. The PowerPoint presentation took approximately 15 minutes. In the second phase, participants moved into groups and had 20 minutes to come up with high-level ideas for how AR could be used to support students learning Japanese. They were instructed to think about very high-level solutions to some of the common struggles they noted in their pre-workshop survey, not just what is technologically possible. Participants received writing utensils, a small whiteboard and associated writing tools to help them document and sketch their ideas. The workshop concluded with a 20-minute discussion between the researchers and the participants, participants discussing their ideas and collectively narrowing to four potential applications. Video recordings were used to generate observational notes and a transcript of the workshop. We analyzed our survey data and transcripts using the data analysis software NVivo. This data was analyzed using an inductive approach – a ground-up approach where codes are derived from the data [10]. Analysis was conducted independently by the lead researcher and research assistant. Results were then compared to identify any key areas of disagreement and resolve codes if needed. Codes were organized into thematic areas [8]. Results of this analysis are shown in Figure 2 below. Following the workshop, the research team prototyped the four applications using Snap Lens Studio, a free desktop application that launched in 2017 and supports the creation of simple

Reference	Year	Focus	Platform	Augmenting
[1]	2021	Kanji	Mobile	Flashcards
[12]	2018	Kanji	Mobile	Tangibles (T-AR)
[13]	2020	Compound verbs	Mobile	Flashcards
[15]	2015	Kanji (radicals)	Mobile	Tangibles (T-AR)
[48]	2021	Kanji	Mobile	Flashcards
[17]	2009	Kanji	Augmented tabletop	Radical cards
[20]	2020	Kanji	Mobile	Flashcards
[45]	2022	Kanji (hiragana + katakana)	Mobile	Flashcards
[30]	2018	Kanji	Mobile	Cards
[35]	2022	Vocabulary, hiragana	Mobile	Book
[37]	2024	Vocabulary, Kanji	Mobile	Location (campus)
[43]	2018	Kanji	Mobile	Mixed Reality
[44]	2003	Kanji	Mobile	Flashcards
[46]	2022	Vocabulary, grammar	HMD	Mixed Reality
[49]	2018	Hiragana	Mobile	Unspecified

Table 1: Overview of related work describing AR applications to support Japanese language learning.

AR projects, called “Lenses” [40]. Due to its lower learning curve, Lens Studio has been utilized as an application to support the rapid prototyping of AR projects [5] and is an especially useful tool for the creation of medium-fidelity prototypes [38]. Lens Studio is able to support functionality for AR prototypes including marker tracking, face tracking, surface tracking and interactive scripting. Any desired features that were not supported by Lens Studio were implemented via the Wizard of Oz paradigm [9]. The results of our workshop and details on the four applications are discussed in the following section.

4 Results and Discussion

4.1 Workshop Outcomes

We conducted a thematic analysis of the final question in our survey and our workshop transcription. The results of this analysis are displayed in Figure 2. The codes were grouped into three themes: struggles for learners, pedagogy struggles and the features of AR. The first theme captures codes that reflect the struggles from the learner’s perspective. Struggles related to the memorization of specific types of content (e.g., grammatical rules, kana or kanji) or gaining proficiency in specific language skills (e.g., listening, mastering keigo). Participants noted that the students focused on content mastery and increased related to reading, writing, speaking, and listening. The mastery of kanji and keigo was considered especially challenging by participants. The second theme includes codes that reflect the struggles from the instructor’s perspective. These include issues such as teaching the importance of using registers (e.g., levels of politeness), motivation and learner perceptions of self-efficacy. For example, with regard to registers, P05 describes a common scenario in which a Japanese teacher addresses their class using formal grammar (e.g., “*ohayou gozaimasu*” = “good morning”) and students reply informally (e.g., “*ohayou sensei!*” “good morning, teacher”). In this example, the levels of politeness have been reversed per socio-cultural expectations in Japanese [28]. While the teacher used the polite register for their students, the students

had responded informally. P07 spoke about the need for students to practice frequently throughout the week outside of the classroom beyond simply completing homework assignments. All participants agreed that practicing in short, frequent bursts is highly beneficial for language learning. They wondered if AR could motivate students by providing exciting ways to engage in course content. P04 talked about how students learn from dialogues in textbooks. The printed dialogue in Japanese language textbooks is often accompanied by multimedia content (e.g., CDs, audio recordings) where students can hear conversations performed by fluent speakers. P04 noted that these are usually performed by Japanese native speakers, but when featuring a fluent foreigner, students more readily felt that they too can one day be fluent. P04 felt it reduced feelings of “impostor syndrome” as students advanced through their studies.

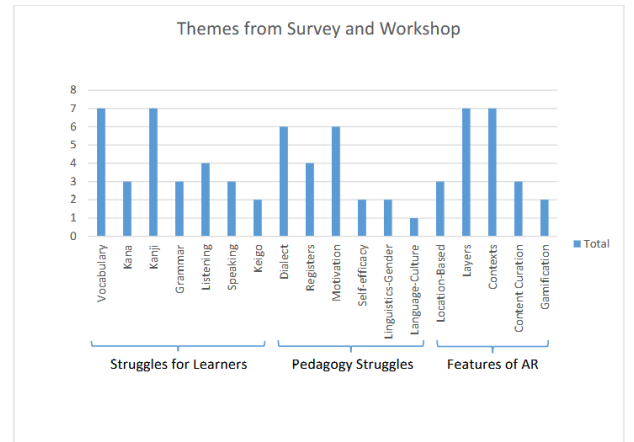


Figure 2: Visualization of themes from analysis of participant surveys and workshop transcripts.

Lastly, our participants noted the issue of standard Japanese, associated with the Tokyo dialect widely taught in Japanese courses.

P05 asked “What is *standard Japanese*? This is becoming an issue in Japanese education.” P01 and P02 presented us with examples where different regions have the same grammatical structure, but different cadence in how they speak. Japanese grammar or words may differ based on regional dialect. Our participants wondered if AR could help present this information to learners in a way that helps them realize this aspect of the Japanese language. The third and final theme captures codes that reflect discussions of how the affordances of AR would be explored to support Japanese language learning. Collaboratively, we discussed the ways in which AR could be used to add layers of content to existing study materials, how other contexts and spaces relevant to the learner could be enhanced with Japanese language content via AR and encourage learners to learn in short bursts. From these discussions, four medium-fidelity prototypes were created in Snap Lens Studio. We present these in the following subsection.

4.2 Prototypes



Figure 3: From left to right (a) photo of the Location-based wayfinding application, (b) a screenshot of a scannable map of Japan that introduces dialects with audio features, (c) a screenshot of an AR application that links audio and written cues to a collection of Japanese restaurant menus and (d) a screenshot of an AR application which adds video content to scannable pages.

4.2.1 Location-based Wayfinding Application. One result of our workshop was the interest that our participants expressed in exploring how AR could encourage multiple, short bursts of practice outside of the context of the classroom or homework assignments. In particular, participants were interested in how the location-based affordances of AR could be utilized to present learners with such opportunities. To this end, we created a design for a location-based wayfinding app based on the unit that introduces wayfinding and giving directions in Japanese. This lesson is introduced as part of Chapter 6 in the Japanese language textbook “Nihongo de Hanasou” as part of the unit on shopping [42]. In this unit, students learn how to follow walking directions by following hand-drawn street maps in the book. In the AR prototype, this lesson is transcribed to the context of wayfinding on the student’s own campus. Students follow printed directions on the screen in Japanese, accompanied by the corresponding spoken audio cues and arrive at specific locations on campus (e.g., the cafeteria, the library, etc.). Gamified mechanics tracking frequency of use and the number of successful navigation

events could be used to further motivate learners extrinsically via an achievement system or class wide leaderboard.

4.2.2 Map to Introduce Dialects. As noted above, P05 initiated discussions around the concept of “standard Japanese” and Japanese dialects. Regional dialects were noted as potentially using the same grammar as “standard Japanese”, but with a different cadence or intonation or having entirely different grammatical structures and words. Our participants wondered if AR could present this information in a way that helped them appreciate this aspect of the Japanese language. This prompted us to design an AR application that would allow learners to scan a map of Japanese regions by dialect and learn more about these differences via textual and audio examples.

Although there are some interactive maps online which present different dialects in Japanese, at this time there is no AR equivalent. While our prototype was built with only simple dialect examples for a few major regions in Japan, the prototype could easily be expanded to a larger map capable of triggering more complex AR content. Continuing this work with Japanese language experts, we could further add historical and cultural contexts to the AR content as well. Participants in our group hailed from many distinct regions of Japan and were very animated when sharing some of the more noteworthy differences between their dialects. Mobilizing a co-design approach in future development of this prototype would allow the community of teachers to build personal narratives into the AR content, potentially strengthening their bonds with their students.

4.2.3 AR Japanese Restaurant Menus. Researchers have noted the importance of integrating authentic materials into the classroom to diversify language learning [31]. In our workshop, participants were interested in the ways in which AR could be used to enhance such materials to further support and engage learners. Our prototype recognizes menu items on a variety of authentic Japanese restaurant menus and could provide learners with additional supports, such as audio cues on demand via AR. Learners could then modify settings to their preference. Some studies indicated that AR can improve the perception of the restaurant experience [3].

Using AR for gastronomy has also been realized, with multiple applications and services available. 3D Menu collaborates with existing restaurants to provide such an experience. The AR menus are web-based and can be accessed via QR code. 3D Menu gives users the ability to interact with each dish on the menu in 3D, exploring each item from every angle before deciding their order. The restaurant has the ability to customize menu branding and track analytics. If more AR menus are produced with these capabilities, there could be higher demand for AR Japanese restaurant menus where users learn kanji characters in the process as well.

However, while 3D Menu is a promising service, it is not an AR translation tool in the context of ordering in another language. While it provides the opportunity for restaurant customers to observe in greater detail what they are ordering, it does not provide linguistic tools to study a second language, as its purpose is specifically for the restaurant industry with the managerial side of the restaurant in mind. Our prototype focuses on the perspective of a secondary language learner, ensuring that through the application they would have access to guides on how to order an item in a

different language, not just learning what they are ordering. By providing learners with an AR application designed around specific pedagogical goals, we can provide learners with a unique opportunity to interact with menus while having access to audio cues on-demand. Furthermore, by building this prototype on a variety of real Japanese menus - many of which utilize stylized Japanese fonts not used in standard Japanese textbooks - it provides students with a more authentic language practice experience.

4.2.4 Content in Japanese Language Textbooks. For the last prototype, we envisioned ways in which AR could enhance Japanese language textbooks. We considered two possibilities, one version where teachers could curate content and link them with tracked markers from scanned sections of the textbook, or a textbook where this content is generated by the publisher and is part of a supplementary mobile app associated to the text. Although the former can be undertaken by language teachers already using applications such as Lens Studio or Vuforia, it may be too complex to widely implement. For the first concept, teachers could be presented with a simple interface to upload a photo or scanned image of a page from their textbook and then pair it with a multimedia file or link to online content. For designer or publisher-led initiatives where companion AR content is included, P04's suggestion of utilizing video content featured fluent non-Japanese individuals to inspire self-efficacy. Here, we present a prototype of the end result for either scenario, featuring video footage of a dialogue in Chapter 1 of our textbook [42].

5 Conclusions and Future Work

In this paper, we presented a co-design workshop exploring the potential applications of Augmented Reality (AR) to support Japanese language learners. Collaborating with a group of seven Japanese language teachers, we identified common struggles faced as observed by our participants and explored potential applications of AR that could address some of these struggles. The research team then created four medium-fidelity prototypes based on the results of the workshop using Snap Lens Studio. Although we were unable to report on the evaluation of fully functional prototypes at this stage in the research, we believe that the results of our study demonstrate the benefits of a collaborative approach to designing AR applications to support Japanese language learners. Specifically, we note how the potential target areas for learning support identified by our participants differs from the focus of many of the AR applications generated previously by researchers. While most of the other AR applications supported vocabulary, kanji, and grammar, insights from our participants identified other struggles beyond "content" that tend to represent issues for learners.

In our future work, we intend to continue iterating on the design of these and future prototypes with other Japanese language instructors. We also plan to fully develop higher fidelity versions of two or more of these prototypes using Unity/Vuforia and evaluate them with language learners, mobilizing pre and post-tests to capture learning, along with observations on their usability and user experience. Furthermore, we are interested in exploring the differences between closed system applications which are designed to exclusively work with specific textbooks/contexts as well as systems which allow Japanese teachers to design their own lessons

with augmented reality. We believe these different approaches each offer their own strengths and weaknesses and can make meaningful contributions to HCI and language pedagogy.

Acknowledgments

We would like to thank our participants for sharing their time and stories with us. Without their enthusiastic participation, this research would not be possible.

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